# MA1522 Linear Algebra for Computing

## Lesson 5: LU Factorization and Determinants

# Submission

- You only need to submit your attempts at those labelled **Question 1** to **Question 3**.
- If you have written your answer on A4 paper, scan or take pictures of your work (make sure the images can be read clearly). Merge all your images into one PDF file, in page order.
- If you have written or typed your answer digitally, print it into a PDF file.
- Name the PDF file by `StudentNo_lesson5` (e.g. `A1234567Z_lesson5.pdf`).
- Submit the PDF file to Canvas assignments, **Participation(L1): Lesson 5**.
- Deadline: today, **12:30 PM**.
- If you do not know how to upload your submission, come to ask me after class.

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Basic rules:

- Please (mainly) ask the questions related to the course.

# Section 2.7 LU Factorization

A square matrix $L$ is **unit lower triangular** if it is lower triangular with every diagonal entry equal to 1. An **LU factorization** of an $m \times n$ matrix $A$ is a decomposition

$$A = LU, \qquad L \text{ unit lower triangular}, \qquad U \text{ a row-echelon form of } A.$$

**The algorithm.** Suppose $A$ can be reduced to a row-echelon form $U$ using only operations $R_i + cR_j$ with $i > j$ — that is, adding a multiple of a higher row to a lower one, never a swap and never a scaling. Writing $E_l$ for the elementary matrix of $r_l$,

$$E_k \cdots E_1 A = U \quad \Longrightarrow \quad A = E_1^{-1} E_2^{-1} \cdots E_k^{-1} U = LU.$$

Each $E_l^{-1}$ is unit lower triangular, and so is any product of such matrices — so $L$ is too.

# Example 1

True or false?

1. Every matrix has an LU factorization.
2. If $A$ has an LU factorization, then it is unique.

(1) **False.** Take $P = \left(\begin{smallmatrix} 0 & 1 \\ 1 & 0 \end{smallmatrix}\right)$. If $P = LU$ then, $L$ being unit lower triangular, the first column of $LU$ is $\left(u_{11},\, l_{21}u_{11}\right)^T$. The $(1,1)$ entry of $P$ forces $u_{11} = 0$, and then the $(2,1)$ entry is $l_{21}u_{11} = 0 \neq 1$. Reducing $P$ needs a row swap, which the algorithm may not use.

(2) **False.** Row 2 of $U$ below is zero, so $t$ multiplies nothing: for every $t \in \mathbb{R}$,

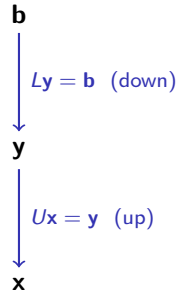
$$\begin{pmatrix} 1 & 1 & 1 \\ 2 & 2 & 2 \\ 3 & 3 & 3 \end{pmatrix} = \begin{pmatrix} 1 & 0 & 0 \\ 2 & 1 & 0 \\ 3 & t & 1 \end{pmatrix} \begin{pmatrix} 1 & 1 & 1 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}.$$

# Section 2.7 Why Bother: Solving $A\mathbf{x} = \mathbf{b}$

With $A = LU$, the system $A\mathbf{x} = \mathbf{b}$ becomes $LU\mathbf{x} = \mathbf{b}$. Put $\mathbf{y} = U\mathbf{x}$ and do it in two easy halves:

- solve $L\mathbf{y} = \mathbf{b}$ — $L$ is unit lower triangular, so read $y_1, y_2, \ldots$ off downwards (forward substitution);
- solve $U\mathbf{x} = \mathbf{y}$ — $U$ is in row-echelon form, so read $x_n, x_{n-1}, \ldots$ off upwards (back substitution).

Neither half needs any row reduction, and $L$, $U$ are computed *once*: a second $\mathbf{b}$ costs only two more substitutions.



# Question 1

Let

$$A = \begin{pmatrix} 2 & 1 & -1 \\ 4 & 3 & 1 \\ -2 & 2 & 10 \end{pmatrix}, \qquad \mathbf{b}_1 = \begin{pmatrix} 2 \\ 8 \\ 10 \end{pmatrix}, \qquad \mathbf{b}_2 = \begin{pmatrix} 2 \\ 6 \\ 0 \end{pmatrix}.$$

1. Find an LU factorization of $A$.

2. Hence solve $A\mathbf{x} = \mathbf{b}_1$ and $A\mathbf{x} = \mathbf{b}_2$.

# Question 1 — solution: part (1)

Reduce, using only $R_i + cR_j$ with $i > j$:

$$\begin{pmatrix} 2 & 1 & -1 \\ 4 & 3 & 1 \\ -2 & 2 & 10 \end{pmatrix} \xrightarrow[R_3+R_1]{R_2-2R_1} \begin{pmatrix} 2 & 1 & -1 \\ 0 & 1 & 3 \\ 0 & 3 & 9 \end{pmatrix} \xrightarrow{R_3-3R_2} \begin{pmatrix} 2 & 1 & -1 \\ 0 & 1 & 3 \\ 0 & 0 & 0 \end{pmatrix} = U.$$

Three operations, three entries of $L$ — each the multiplier with the sign flipped:

$$\underbrace{R_2 - 2R_1}_{L_{21}=2}, \qquad \underbrace{R_3 + R_1}_{L_{31}=-1}, \qquad \underbrace{R_3 - 3R_2}_{L_{32}=3},$$

$$A = LU = \begin{pmatrix} 1 & 0 & 0 \\ 2 & 1 & 0 \\ -1 & 3 & 1 \end{pmatrix} \begin{pmatrix} 2 & 1 & -1 \\ 0 & 1 & 3 \\ 0 & 0 & 0 \end{pmatrix}.$$

Check: row 3 of $LU$ is $-1 \cdot (2, 1, -1) + 3 \cdot (0, 1, 3) + (0, 0, 0) = (-2, 2, 10)$. ✓

# Question 1 — solution: part (2)

Forward: $L\mathbf{y} = \mathbf{b}_1$, read downwards — $y_1 = 2$; $2y_1 + y_2 = 8$ so $y_2 = 4$; $-y_1 + 3y_2 + y_3 = 10$ so $y_3 = 0$. Back: $U\mathbf{x} = \mathbf{y}$ — row 3 reads $0 = 0$, so $x_3$ is free. Put $x_3 = s$; then $x_2 + 3s = 4$ and $2x_1 + x_2 - s = 2$.

$$\mathbf{x} = \begin{pmatrix} -1 \\ 4 \\ 0 \end{pmatrix} + s \begin{pmatrix} 2 \\ -3 \\ 1 \end{pmatrix}, \qquad s \in \mathbb{R} \quad \text{— infinitely many solutions.}$$

Same $L$, same $U$ for $\mathbf{b}_2$; only the substitutions are redone. Now $L\mathbf{y} = \mathbf{b}_2$ gives $\mathbf{y} = (2, 2, -4)^T$, and row 3 of $U\mathbf{x} = \mathbf{y}$ reads

$$0 = -4 \quad \text{— no solution.}$$

The zero row of $U$ decides both: it is why $\mathbf{b}_1$ has a free variable and why $\mathbf{b}_2$ has nothing at all. □

# Challenge

Write a program that decides whether the algorithm of Section 2.7 produces an LU factorization of a matrix $A$, and outputs $L$ and $U$ when it does.

# Section 2.8 Determinant by Cofactor Expansion

For $A$ square, $\det(A)$ (also written $|A|$) is defined by induction: $\det(a) = a$, $\det\left(\begin{smallmatrix} a & b \\ c & d \end{smallmatrix}\right) = ad - bc$, and in general through **minors** and **cofactors**:

$$M_{ij} = \text{delete row } i \text{ and column } j \text{ of } A, \qquad A_{ij} = (-1)^{i+j} \det(M_{ij}).$$

The signs $(-1)^{i+j}$ in place:

$$\begin{pmatrix} + & - & + & \cdots \\ - & + & - & \cdots \\ + & - & + & \cdots \\ \vdots & \vdots & \vdots & \ddots \end{pmatrix}$$

## Cofactor expansion

For any row $i$, and for any column $j$,

$$\det(A) = \sum_{k=1}^{n} a_{ik} A_{ik} = \sum_{k=1}^{n} a_{kj} A_{kj}.$$

Every choice gives the same number.

- $\det(A^T) = \det(A)$ — so anything true of rows is true of columns.
- If $A$ is triangular, $\det(A) = a_{11} a_{22} \cdots a_{nn}$.

# Question 2

Let $t \in \mathbb{R}$ and

$$A = \begin{pmatrix} 2 & 0 & 1 & 3 \\ 1 & 0 & 4 & 2 \\ -1 & 3 & 2 & 0 \\ 0 & 0 & -2 & t \end{pmatrix}.$$

Compute $\det(A)$ by cofactor expansion, and find the value of $t$ for which $\det(A) = 0$.

# Question 2 — solution

Column 2 has a single nonzero entry, $a_{32} = 3$, so the expansion along it has only one surviving term — whatever $t$ is:

$$\det(A) = a_{32}A_{32} = 3 \cdot (-1)^{3+2} \det \begin{pmatrix} 2 & 1 & 3 \\ 1 & 4 & 2 \\ 0 & -2 & t \end{pmatrix}$$

$$= -3\Big(2(4t+4) - 1(t-0) + 3(-2-0)\Big) = -3(7t+2) = -21t - 6.$$

So $\det(A) = 0$ exactly when

$$t = -\tfrac{2}{7}.$$

Next lesson we find out what is special about a matrix whose determinant is zero. □

# Section 2.9 Determinant by Reduction

Cofactor expansion on an $n \times n$ matrix costs about $n!$ multiplications. Row reduction costs about $n^3$. So: reduce, and track what the operations do.

## One row operation

$$A \xrightarrow{R_i + cR_j} B \implies \det(B) = \det(A)$$

$$A \xrightarrow{cR_i} B \implies \det(B) = c\det(A)$$

$$A \xrightarrow{R_i \leftrightarrow R_j} B \implies \det(B) = -\det(A)$$

## The whole reduction

If $A \xrightarrow{r_1} \cdots \xrightarrow{r_k} U$ with $U$ a row-echelon form with diagonal $d_1, \ldots, d_n$, then

$$\det(A) = \frac{d_1 d_2 \cdots d_n}{\det(E_k) \cdots \det(E_1)}.$$

# Question 3

Take the same matrix as Question 2:

$$A = \begin{pmatrix} 2 & 0 & 1 & 3 \\ 1 & 0 & 4 & 2 \\ -1 & 3 & 2 & 0 \\ 0 & 0 & -2 & t \end{pmatrix}.$$

Compute $\det(A)$ by row reduction. Your answer must agree with Question 2.

# Question 3 — solution

Swap first, to get a leading 1; the route below keeps every entry free of fractions:

$$\begin{pmatrix} 2 & 0 & 1 & 3 \\ 1 & 0 & 4 & 2 \\ -1 & 3 & 2 & 0 \\ 0 & 0 & -2 & t \end{pmatrix} \xrightarrow{R_1 \leftrightarrow R_2} \begin{pmatrix} 1 & 0 & 4 & 2 \\ 2 & 0 & 1 & 3 \\ -1 & 3 & 2 & 0 \\ 0 & 0 & -2 & t \end{pmatrix} \xrightarrow[R_3+R_1]{R_2-2R_1} \begin{pmatrix} 1 & 0 & 4 & 2 \\ 0 & 0 & -7 & -1 \\ 0 & 3 & 6 & 2 \\ 0 & 0 & -2 & t \end{pmatrix} \xrightarrow{R_2 \leftrightarrow R_3} \begin{pmatrix} 1 & 0 & 4 & 2 \\ 0 & 3 & 6 & 2 \\ 0 & 0 & -7 & -1 \\ 0 & 0 & -2 & t \end{pmatrix}$$

$$\xrightarrow{R_3-3R_4} \begin{pmatrix} 1 & 0 & 4 & 2 \\ 0 & 3 & 6 & 2 \\ 0 & 0 & -1 & -1-3t \\ 0 & 0 & -2 & t \end{pmatrix} \xrightarrow{R_4-2R_3} \begin{pmatrix} 1 & 0 & 4 & 2 \\ 0 & 3 & 6 & 2 \\ 0 & 0 & -1 & -1-3t \\ 0 & 0 & 0 & 7t+2 \end{pmatrix} = U.$$

Diagonal product $1 \cdot 3 \cdot (-1) \cdot (7t+2) = -3(7t+2) = -21t - 6$; two swaps contribute $(-1)(-1) = 1$; the four $R_i + cR_j$ contribute nothing. So $\det(A) = -21t - 6$, exactly as in Question 2, for every $t$. ✓ □

# Challenge

Let $A$ be a square matrix of order $n$. Use both definitions of the determinant — cofactor expansion (Section 2.8) and reduction (Section 2.9) — to explain why

$$\det(-A) = (-1)^n \det(A).$$

# Next lesson

- **Sections 2.8–2.10 Determinants and Their Properties**

Please watch the video clips for these sections beforehand.

If you do not know how to upload your submission, come to ask me after class.